Input Sampler

Parameters

- DATA_WIDTH.
- PARTIAL_SIZE.
- DATA_IN_SIZE = DATA_WIDTH * PARTIAL_SIZE
- NUMBER_OF_PARTIALS_PER_OUTPUT = 2.
- DATA_OUT_SIZE = DATA_WIDTH * NUMBER_OF_PARTIALS_PER_OUTPUT

Input Signals

Signal	Number of bits	Description
rst_n	1	When asserted, delete all the registers and the outputs are = 0
sample_data	1	When high, means the data_in is valid and should be used otherwise the data_in is discarded
data_in	DATA_IN_SIZE	data_in when sample_data is high, means the data_in is valid and should be used otherwise the data_in is discarded

Output Signals

Signal	Number of bits	Description
data_out	DATA_OUT_SIZE	Contains all the data sent before concatenated with the recent data_in while the oldest is the most significant bit and the data_in is the least significant bit
valid_out	1	When the counter becomes = NUMBER_OF_PARTIALS_PER_OUTPUT

Notes

You should make a counter that counts the number that the sample_data is asserted and save each data_in when the sample_data is asserted to a reg ([DATA_SIZE – 1 : 0] temp [NUMBER_OF_PARTIALS_PER_OUTPUT – 2 : 0]) until the counter reaches the

NUMBER_OF_PARTIALS_PER_OUTPUT and in the same clock cycle you should concatenate all the data in the reg with the data_in and put them in the output without missing a beat.